

DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

Anonymous authors
Anonymous submission

ABSTRACT

Parametric imitation learning through behavior cloning remains susceptible to poor generalization on out-of-distribution states, largely because small deployment-time errors compound as closed-loop trajectories depart from the demonstrated state distribution. We show that this vulnerability can be mitigated by reusing the training corpus during inference within a semi-parametric, retrieval-based imitation-learning framework. We introduce **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, a semi-parametric retrieval-based approach that reformulates imitation learning around local neighborhood geometry rather than direct global state-to-action regression. Instead of fitting a single global policy in isolation, DARP trains a predictor whose action estimates are conditioned on k -nearest neighbors from expert demonstrations, their associated expert actions, and the relative displacement vectors between retrieved neighbor states and query states. DARP preserves the assumptions of standard behavior cloning: it requires neither additional data collection, nor online expert correction, nor task-specific structural knowledge. Across diverse domains, including continuous control and robotic manipulation, and across representations ranging from low-dimensional states to high-dimensional visual features, DARP yields consistent 15-46% performance improvements over standard behavior cloning. Code and demos will be released upon acceptance.

1 INTRODUCTION

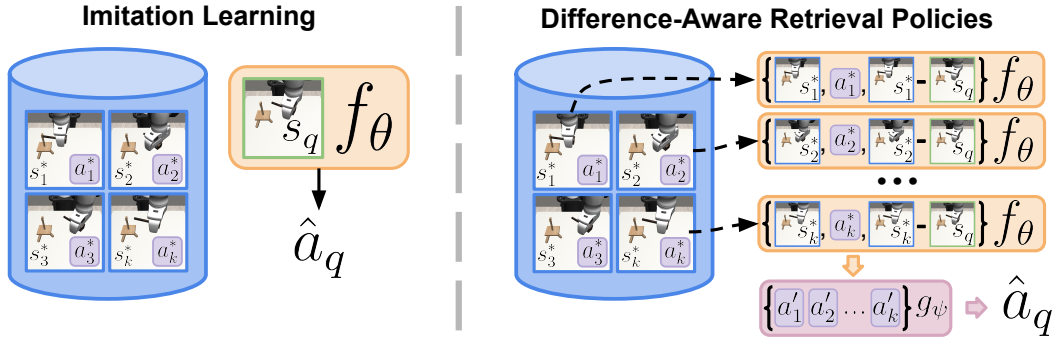


Figure 1: **Overview of DARP:** In contrast to standard BC (left), DARP (right) employs a retrieval-based reparameterization organized around difference vectors between query states and retrieved neighbors. Whereas standard behavior cloning uses expert state-action pairs only during training and discards the dataset at inference time, DARP retains the corpus and retrieves a local neighborhood of expert state-action pairs around each query point s_q .

Imitation learning through behavior cloning (BC) (Pomerleau, 1991) has enabled robotic systems to acquire complex, dexterous behaviors from expert demonstrations (Zhao et al., 2023; Chi et al., 2024; Black et al., 2024; Chung et al., 2014). Nevertheless, despite its conceptual and algorithmic simplicity, BC often exhibits pronounced brittleness in deployment, particularly on long-horizon tasks (Ross et al., 2011). The fundamental difficulty is *covariate shift*: small action errors accumulate

during rollouts, progressively transporting the agent into states that are poorly represented in the demonstration corpus (Spencer et al., 2021; Ross et al., 2011). Within these out-of-distribution regions, BC policies may become highly unstable, producing unreliable, high-variance behavior that frequently culminates in task failure.

This failure mode is well understood, and a substantial body of work has proposed mechanisms for attenuating compounding error (Ross et al., 2011; Venkatraman et al., 2015; Ke et al., 2024b; Levine et al., 2020). Yet these methods typically depart from the standard BC setting by requiring simulators, interactive experts, large quantities of sub-optimal data, or strong task-specific structure. Our objective, by contrast, is to remain within the pure BC regime: learning exclusively from expert state-action pairs, without additional supervision, online correction, or auxiliary feedback. This constraint motivates the central question of the paper: *can the variance of BC policies be reduced using only the original demonstration dataset?*

From a statistical perspective, BC minimizes supervised risk only on expert states. This objective controls bias on the training distribution but leaves variance largely unconstrained: in low-density regions of the state space, which are commonly encountered during closed-loop rollouts, the learned policy can oscillate arbitrarily. A natural remedy is to impose *smoothness*, so that nearby states induce similar predicted actions (Kobayashi, 2022; Asadi et al., 2018; Ke et al., 2024a; Chen et al., 2024). Such smoothness discourages spurious fluctuations and thereby improves rollout stability. As discussed further in Section 4, several approaches have explored explicit mechanisms for encouraging this property.

Although these strategies can be effective, each introduces characteristic limitations: augmentation does not guarantee consistency, global priors can obscure legitimately distinct behaviors, temporal penalties operate only along time rather than across state-space neighborhoods, and explicit graph regularizers require additional smoothness hyperparameters whose tuning can be delicate.

A complementary perspective distinguishes global from local learning. “Global” supervised models (Black et al., 2024; Zhao et al., 2023; Chi et al., 2024) attempt to compress the entire demonstration corpus into a single parametric function, a strategy that can become brittle under distribution shift. “Local” methods (Pari et al., 2022; Mansimov & Cho, 2018; Salzberg & Aha, 1994), in contrast, condition their predictions on the dataset’s neighborhood structure, consulting similar states and producing outputs through non-parametric or semi-parametric operations on the empirical distribution of expert behavior. This locality confers robustness by reducing reliance on a single globally extrapolating function; however, it also introduces limitations, since performance depends intrinsically on the distance metric, naive neighborhood averaging can blur distinct actions and represent multimodality poorly, and treating neighbors solely through their absolute states can restrict generalization.

We introduce **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, which integrates the robustness of local methods with the stability of regularized global policy learning. At inference time, rather than predicting an executable action solely through a feedforward evaluation of the current query state, DARP (Fig. 1) retrieves k neighbors from the training corpus and predicts k action candidates conditioned on tuples comprising the neighbor state, its associated expert action, and the difference vector from the query state to that neighbor. These candidates are then aggregated into a final action. In this sense, DARP reparameterizes imitation learning around local difference structure, allowing the model to exploit expert neighborhoods while still learning a flexible parametric correction.

This reparameterization yields two benefits that are both theoretically and empirically consequential. First, we show that enforcing local consistency in the policy corresponds to manifold regularization, or Laplacian smoothing, over the graph induced by expert demonstrations; this regularization reduces variance, promotes local smoothness, and improves closed-loop stability relative to vanilla BC. Second, instead of adding an explicit graph regularizer and tuning an additional coefficient, DARP builds the corresponding smoothing operation into the architecture through retrieval and permutation-invariant aggregation. The resulting method retains the standard imitation-learning objective while implicitly inducing the local consistency that explicit regularization would otherwise impose.

Empirically, DARP produces consistent gains across continuous-control and robotic-manipulation benchmarks. Across MuJoCo locomotion, Robosuite manipulation, RoboCasa kitchen tasks, and real-world FurnitureBench assembly, DARP improves over standard BC by 15-46%, while also scaling to visual representations and multimodal action distributions. Ablations further show that

the difference-vector parameterization and permutation-invariant aggregation are central to these improvements, indicating that the gains do not arise merely from ensembling or retrieval alone but from the particular difference-aware semi-parametric architecture.

Our contributions are threefold. First, we formalize a neighbor-regularized imitation-learning objective and show that it improves the variance, smoothness, and stability properties of BC. Second, we demonstrate that these benefits can be obtained implicitly through a retrieval-augmented architecture, without changing the standard BC objective or introducing a smoothness hyperparameter. Third, we instantiate this principle as DARP and show through extensive experiments that difference-aware retrieval policies substantially outperform standard behavior cloning across state-based, vision-based, multimodal, simulated, and real-world settings.

2 DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

In this work, we instantiate a class of imitation-learning methods designed to synthesize the complementary advantages of “global” parametric policies and “local” neighborhood-based predictors. We propose an architectural reparameterization, coupled with a simple training objective, that preserves the data and supervision requirements of ordinary behavior cloning while yielding substantial theoretical and empirical improvements. This section develops the theoretical motivation from first principles; readers primarily interested in the operational algorithm may proceed to Section 2.5, which gives a self-contained summary of DARP, before turning to the empirical evaluation in Section 3.

We begin by formalizing the problem setting (Section 2.1) and then introduce a regularized variant of imitation learning (Section 2.2) that uses neighborhood structure in the data to improve variance, generalization, and stability. Section 2.3 subsequently shows that the benefits of explicit regularization can be obtained implicitly by modifying the policy *architecture* rather than the loss. Finally, Section 2.4 presents the practical algorithm DARP, which operationalizes these ideas through a semi-parametric retrieval-augmented architecture compatible with modern neural-network and generative-modeling tools.

2.1 PRELIMINARIES: BEHAVIOR CLONING FOR IMITATION LEARNING

We consider the standard imitation-learning setting, formalized as a finite-horizon Markov Decision Process (MDP), $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, P_0\}$, where $\mathcal{S}$ denotes the state space, $\mathcal{A}$ the action space, and P_0 the initial-state distribution. A *policy* maps each state to a distribution over actions, $f_\theta : \mathcal{S} \rightarrow \Delta_{\mathcal{A}}$, with the aim of optimizing task-relevant objectives; for notational economy, we omit the subscript θ when referring to the generic function class f . We assume access to expert human-provided demonstrations $\mathcal{D}^*$, represented as state-action pairs $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$, and use s^* and a^* specifically for states and actions drawn from this expert corpus. Behavior cloning (Pomerleau, 1991) learns f_θ by reducing imitation to supervised learning, namely $\arg \max_{\theta} \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\log(f_\theta(a^* | s^*))]$. Although f_θ may be parameterized by a rich generative family (Lipman et al., 2023; Chi et al., 2024), we initially adopt a Gaussian parameterization for analytical simplicity¹.

2.2 WARM-UP: NEIGHBOR MANIFOLD REGULARIZED IMITATION LEARNING

Although behavior cloning minimizes the supervised imitation loss on expert states sampled from $\mathcal{D}^*$, this pointwise objective by itself imposes no constraint on policy behavior away from the expert-state manifold. Consequently, as deployment errors accumulate, the agent may enter out-of-distribution regions in which a BC policy, particularly when represented by an overparameterized neural network, can behave arbitrarily (Ross & Bagnell, 2010).

To address this limitation, observe that BC constrains function values only at training states and does not explicitly encode the relationships among nearby states and their corresponding actions, thereby neglecting the geometry of the underlying data manifold. We incorporate this information by considering a modified objective with an explicit regularization term that promotes *local consistency*:

¹We show that this can be relaxed in Section 2.4

expert states that are close in the dataset should induce similar policy predictions. This intuition yields the following neighborhood-regularized loss ($\mathcal{L}_{\text{MRIL}}$), in which the standard imitation-learning objective ($\mathcal{L}_{\text{BC}}$) is augmented by a smoothness penalty ($\mathcal{L}_S$) that forces predictions to respect dataset geometry rather than relying exclusively on pointwise supervision.

$$\mathcal{L}_{\text{MRIL}}(f) = \underbrace{\mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)]}_{\text{supervised risk}(\mathcal{L}_{\text{BC}})} + \underbrace{\lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right]}_{\text{smoothness regularizer}(\mathcal{L}_S)}, \quad (1)$$

Here, $\ell(f(s^*), a^*)$ denotes the supervised imitation loss, $\mathcal{N}_k(s^*)$ denotes the k -nearest neighbors of s^* in the expert dataset, and $w_i(s^*)$ are normalized kernel weights determined by state differences, namely $w_i(s^*) \propto K_\Delta\left(\frac{\|s_i^* - s^*\|}{h}\right)$. As summarized below and developed in Appendix A.1.1, this objective instantiates *manifold regularization*, or Laplacian smoothing, by penalizing high-frequency variation of the policy across neighborhoods of expert states. This form of regularization yields provable improvements in variance, stability, and generalization.

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ be the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$. Let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider two estimators trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Then, under the smoothness assumption on f , the following hold:

- (i) Variance reduction: The Laplacian penalty in MRIL acts as a data-dependent Tikhonov regularizer, yielding smaller estimator variance than vanilla BC.
- (ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants with arbitrarily large Lipschitz constants between training states.
- (iii) Policy stability: In a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_S \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^2$$

accumulates error linearly for vanilla BC, but sublinearly for MRIL, since the smoothness regularizer enforces $\|f(s^*) - f(s'^*)\| = O(\|s^* - s'^*\|)$ for neighbors s^*, s'^* .

This suggests that MRIL enjoys strictly better generalization and stability guarantees than BC.

Proof sketch. We provide the complete proof in Appendix A.1.1; here we summarize the main argument. The proof first identifies the smoothness term with a graph-Laplacian penalty on the graph induced by a k -nearest-neighbor (k -NN) affinity matrix whose edge weights are determined by the kernel w_i . It then shows that, as the sample size increases, this graph-Laplacian penalty converges to a weighted Dirichlet energy (Belkin & Niyogi, 2008; Zhou et al., 2003). Minimization of this energy both makes the learned f locally Lipschitz almost everywhere, thereby inducing smoothness and policy stability, and acts as Tikhonov regularization, thereby reducing estimator variance while keeping bias controlled. $\square$

² L_S and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

Conceptually, the smoothness regularizer does more than penalize pairwise disagreement among neighbors: it encourages the learned policy to vary smoothly with respect to the underlying data manifold. In particular, it contracts the local Lipschitz constant of f along directions of high data density $p(s)$, ensuring that small state perturbations produce small and coherent changes in the predicted action. The resulting policy therefore generalizes more reliably on in-distribution (ID) states and extrapolates more systematically to nearby out-of-distribution (OOD) states.

2.3 IMPLICIT MANIFOLD REGULARIZATION VIA IN-CONTEXT ARCHITECTURES

Although the MRIL objective amortizes local learning and thereby improves over vanilla BC, it introduces two important drawbacks. First, it requires a hyperparameter λ whose value mediates the tradeoff between supervised accuracy and smoothness. Second, replacing the standard BC objective with a regularized one may alter the optimization landscape in undesirable ways. These considerations motivate a natural question: *can the benefits conferred by MRIL (Eq. 1) be recovered through architectural design rather than objective modification?*

We therefore introduce a retrieval-based architectural modification that produces an *implicit* manifold-regularization effect (iMRIL) while retaining the standard imitation objective. Through iMRIL, the benefits of Laplacian smoothing associated with MRIL can be obtained under ordinary BC training, as illustrated in Fig. 2, without introducing λ as an additional training hyperparameter. Section 2.4 then builds on this construction to define the practical algorithm DARP.

iMRIL architecture: The central idea behind iMRIL is to relocate the neighborhood-aggregation operation from the loss, as in Eq. 1, into the architecture itself. Rather than training a conventional feedforward predictor $f(s)$ against a neighborhood-regularized smoothness objective, we embed aggregation directly into the parameterization of the action predictor $\hat{f}$ while keeping the objective identical to standard imitation learning. Specifically, iMRIL learns a per-state predictor f_θ such that the neighborhood-aggregated predictor $\hat{f}(s^*) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*)$, evaluated over nearest-neighbor training states $\{s_i^*\}_{i \in \mathcal{N}_k(s^*)}$, accurately reconstructs the corresponding expert action a^* . Under this parameterization, iMRIL optimizes the following training objective:

$$\arg \min_{\theta} \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} \left[\left\| \underbrace{\left(\frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*) \right)}_{\hat{f}(s^*)} - a^* \right\|_2 \right] \quad (2)$$

At deployment time, inference on a novel state s_q proceeds by retrieving the k -NN of s_q from the training set and applying neighborhood aggregation, $\hat{a} = \hat{f}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_\theta(s_i^*)$. As Section 2.4 demonstrates, the precise parameterization of f is empirically consequential and motivates the full DARP architecture.

Intuitively, the action predictor $\hat{f}$ is parameterized as an aggregation of predictions at neighboring training states, $f(s_i^*)$, and the underlying function f is learned through supervision applied after aggregation. Because the supervised object is the post-aggregation predictor, individual f evaluations are implicitly discouraged from becoming arbitrarily non-smooth, thereby conferring the benefits identified in Section 2.2. We formalize this intuition by proving a direct equivalence between iMRIL and Laplacian regularization.

Equivalence between iMRIL and MRIL: Although the full proof of the formal equivalence between MRIL and iMRIL is deferred to Appendix Section A.1, we state the principal result and provide a proof sketch here.

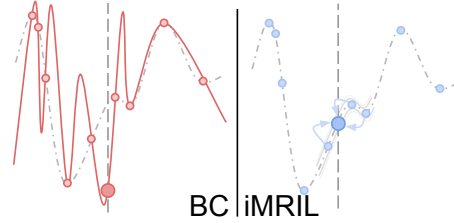


Figure 2: **iMRIL implicitly realizes Laplacian smoothing, reducing variance while enforcing local consistency**, whereas standard BC, absent an explicit smoothness constraint, can admit arbitrarily jagged function approximations.

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). *Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.*

The minimizers of the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) have the following closed form expansions

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

iMRIL's neighbor-aggregation step applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , preserving low-frequency modes while attenuating high-frequency modes. The correspondence between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} therefore shows that iMRIL implements a built-in form of Laplacian smoothing (MRIL) with effective $\lambda \approx 1$ in normalized units. Unlike explicit regularization, this implicit filter introduces no additional hyperparameter to tune.

Proof sketch. Full details appear in Appendix Section A.1.2. The explicit regularizer admits a spectral solution obtained by diagonalizing the k -NN Laplacian, which yields a filter of the form $(1 + \lambda \mu)^{-1}$ on each eigenmode. The implicit objective can be written as neighbor aggregation $\hat{f} = S f$ with $S = D^{-1} A$, the random-walk matrix, which shares the same eigenvectors and applies the fixed filter $1 - \mu$. Both constructions therefore operate as graph low-pass filters: modes with small eigenvalues, corresponding to smooth variation along the data manifold, are largely preserved, whereas modes with large eigenvalues, corresponding to rapid high-variance fluctuation between neighbors, are strongly suppressed. Thus iMRIL performs Laplacian smoothing implicitly, reducing variance and enforcing local consistency without requiring the tuning of λ . $\square$

Importantly, the implicit-Laplacian-smoothing interpretation does not eliminate the need to learn a policy; instead, it constrains the class of post-aggregation functions that the architecture can represent. The neighbor-conditioned network f_θ learns how expert actions change under local perturbations, proposing locally adapted candidate actions for each neighbor. The aggregation operator then reduces variance by smoothing these candidates across the neighborhood. Thus, learning supplies accuracy by correcting local bias, while aggregation supplies stability by controlling variance.

2.4 DIFFERENCE-AWARE RETRIEVAL POLICIES: A PRACTICAL INSTANTIATION OF iMRIL FOR IMITATION LEARNING

Having established the conceptual framework of iMRIL, we now instantiate a practical algorithm for large-scale imitation learning. Building on the objective in Eq. 2, we specify a particular parameterization and neighborhood-aggregation mechanism that together yield strong empirical performance.

2.4.1 DIFFERENCE-BASED PARAMETERIZATION OF f_θ

The objective in Eq. 2 leaves the parameterization and input representation of f_θ underdetermined. We observe that, if neighborhood aggregation is to be effective, the model should learn how expert actions vary under local perturbations. This suggests that f_θ should condition on *differences* between the query state and each neighbor state, thereby enabling locally adapted action proposals. In **Difference-Aware Retrieval Policies (DARP)**, rather than parameterizing f_θ only as $f_\theta(s_i^*)$, we provide the model with the optimal neighbor action a_i^* and the *difference* between the query and neighbor states, $\Delta s_i = s_i^* - s_q$; the predictor then produces an action candidate a_i' for query state s_q and neighbor (s_i^*, a_i^*) via $a_i' = f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)$.

Let $\mathcal{N}_k(s_q)$ denote the index set of the k -nearest neighbors retrieved according to a distance function $d(s_q, s_i^*)$.³ Given these retrieved neighbors, DARP predicts an action for any query state s_q by applying the neighborhood-aggregation operation introduced in Section 2.3

³In our work we use the Euclidean distance in a pre-trained embedding space, although other neighborhood functions are also applicable. We refer the reader to Appendix Section A.2.1 for a thorough discussion of design decisions in constructing neighborhood sets via retrieval.

$$\hat{a}_q = f_{\text{DARP}}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i \quad (3)$$

$$= \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_{\theta}(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q). \quad (4)$$

During training, this architecture induces the following direct imitation-learning objective over the expert dataset $\mathcal{D}^*$:

$$\arg \min_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\|f_{\text{DARP}}(s_q^*) - a_q^*\|^2] \quad (5)$$

The parameters of f_{θ} are optimized by minimizing the discrepancy between the predicted action $\hat{a}_q$ and the expert action a_q^* . Because the objective is deliberately simple, f_{θ} can be instantiated with a range of function approximators; in our experiments, we use standard feedforward or convolutional neural networks. As shown in Section 3, the difference-based parameterization is critical for performance. At inference time, executable actions are produced by retrieving the k -NN set and evaluating the neighborhood-aggregation operation in Eq. 3.

2.4.2 GOING BEYOND LINEAR AGGREGATION

Although the neighborhood-aggregation mechanism has so far been presented as simple averaging, $\hat{a}_q = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i$, this choice is only one member of a broader family of permutation-invariant aggregation functions $g_{\psi}(\{a'_i\}_{i \in \mathcal{N}_k(s_q)})$. For example, g_{ψ} could be parameterized by more expressive set-compatible neural architectures such as the set transformer (Lee et al., 2019) or DeepSets (Zaheer et al., 2017). This observation generalizes the prediction model in Eq. 3 to $\hat{a}_q = g_{\psi}(\{f_{\theta}(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$. Beyond improved expressivity, replacing averaging with a parametric aggregation model g_{ψ} permits richer action distributions, such as Gaussian mixture models (Pignat & Calinon, 2019) or diffusion models (Chi et al., 2024), rather than the Gaussian distribution implicit in the L_2 -regression objective of Eq. 5. Instead of predicting $\hat{a}_q$ directly, DARP can predict parameters α of an action distribution $p(a_q; \alpha)$, including, for example, Gaussian-mixture means, covariances, and weights, or the score function of a diffusion model. This extension enables maximum-likelihood training of multimodal action distributions rather than restricting the model to unimodal L_2 regression:

$$\arg \max_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\log p(a_q^*; \alpha_{\theta}(s_q^*))], \text{ where } \alpha_{\theta}(s_q^*) = g_{\psi}(\{f_{\theta}(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q^*)\}_{i \in \mathcal{N}_k(s_q^*)}) \quad (6)$$

For a query state s_q , inference then proceeds by constructing $\alpha_{\theta}(s_q) = g_{\psi}(\{f_{\theta}(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$ from the neighbors retrieved at test time and sampling $\hat{a}_q \sim p(\cdot; \alpha_{\theta}(s_q))$. Detailed training pseudocode is provided in Appendix Section A.3.

2.5 SUMMARY

We have shown that promoting local consistency while fitting a behavior-cloning policy corresponds to manifold regularization, or Laplacian smoothing, and that this structure provably improves variance, smoothness, and stability relative to standard BC (Theorem 1). Rather than inserting an explicit regularization term, together with its associated hyperparameter λ , into the loss, we show that smoothing can be induced implicitly by incorporating neighborhood aggregation directly into the policy architecture (Theorem 2). The resulting algorithm is therefore as follows: during training, for each expert state-action pair (s_q^*, a_q^*) , DARP retrieves the k -nearest neighbor states from $\mathcal{D}^*$, computes difference vectors $\Delta s_i = s_i^* - s_q^*$ from each neighbor to the query state, maps each tuple $(s_i^*, a_i^*, \Delta s_i)$ through a network f_{θ} to obtain candidate actions, and aggregates those candidates with a permutation-invariant function g_{ψ} to predict the final action. The entire procedure is trained using only the standard imitation-learning objective.

3 EXPERIMENTAL EVALUATION

We evaluate DARP by addressing three questions: **Q1**: Can DARP consistently outperform standard behavior cloning?, **Q2**: Can DARP accommodate more complex state representations and action distributions?, and **Q3**: Which architectural components account for DARP’s performance gains? Our experiments span multiple domains and representation regimes, including low-dimensional states, high-dimensional visual features, and heterogeneous action representations. The evaluation covers continuous-control benchmarks in MuJoCo, robotic-manipulation tasks in Robosuite and RoboCasa, and deliberately discontinuous environments that stress-test the neighbor-based formulation.

3.1 BASELINE COMPARISONS AND TASK DESCRIPTIONS

MuJoCo Tasks: The MuJoCo (Todorov et al., 2012; Fu et al., 2020) tasks require controlling legged agents with different morphologies so that they achieve stable forward locomotion on a flat plane. The suite includes Hopper, a single-legged hopping robot; Walker, a bipedal humanoid; Ant, a quadruped; and HalfCheetah, a bipedal runner.

Robosuite Tasks: The Robosuite (Zhu et al., 2020) tasks involve single-arm object manipulation. Stack requires placing a smaller cube on top of a larger cube; Threading requires manipulating a thin, needle-like tool and inserting it into a small ring; and Square Peg requires moving a square wooden block with a central hole onto a square peg.

RoboCasa Tasks: The RoboCasa (Nasiriany et al., 2024) tasks require a single robotic arm to manipulate objects in randomized kitchen scenes. Drawer requires closing an open drawer, Door requires closing an open microwave door, and Stove requires rotating a knob to turn off a stove burner.

Baseline Comparisons: We compare DARP with several baselines and ablations: (1) **R&P (Sridhar et al., 2025)**: directly executes the action associated with the nearest neighbor; (2) **LWR (Pari et al., 2022)**: performs locally weighted regression over retrieved neighbors; (3) **BC**: denotes standard parametric behavior cloning; (4) **REGENT (Sridhar et al., 2025)**: denotes a transformer-based in-context learning method conditioned on retrieved neighbors; and (5) **MRIL**: denotes the explicitly smoothed variant of DARP introduced in Section 2.2.

3.2 CAN DARP CONSISTENTLY OUTPERFORM STANDARD BEHAVIOR CLONING? (Q1)

We first test DARP’s central hypothesis on tasks with low-dimensional state representations, where inter-state distance metrics are well defined and comparatively interpretable. This evaluation spans MuJoCo locomotion as well as Robosuite and RoboCasa manipulation tasks, with manipulation data generated using MimicGen (Mandlekar et al., 2023). In these experiments, the aggregation function g is instantiated as the simple average of all neighbor-conditioned action predictions a' .

Method	Hopper	Ant	Walker	HalfCheetah
R&P (Sridhar et al., 2025)	711.82 $\pm$ 85.63	-305.97 $\pm$ 76.42	419.18 $\pm$ 50.21	-178.64 $\pm$ 29.75
LWR (Pari et al., 2022)	1703.78 $\pm$ 245.95	846.59 $\pm$ 216.06	1484.91 $\pm$ 356.54	1945.82 $\pm$ 567.26
BC	2313.65 $\pm$ 203.75	2376.20 $\pm$ 339.43	2658.40 $\pm$ 274.08	1063.23 $\pm$ 371.08
REGENT (Sridhar et al., 2025)	1819.39 $\pm$ 186.24	-302.10 $\pm$ 146.67	507.01 $\pm$ 76.10	169.85 $\pm$ 63.10
MRIL	2793.63 $\pm$ 156.41	3869.08 $\pm$ 241.00	4370.96 $\pm$ 168.13	701.58 $\pm$ 195.08
DARP	3545.57 $\pm$ 3.54	4383.28 $\pm$ 266.37	4894.01 $\pm$ 75.12	5515.41 $\pm$ 841.33
DARP Set Transformer	2965.86 $\pm$ 103.08	4063.79 $\pm$ 218.80	4752.42 $\pm$ 109.23	3417.85 $\pm$ 764.57

Table 1: **Both DARP and DARP Set Transformer outperform the competing approaches across all domains.** Performance comparison between DARP, BC, and additional baselines on MuJoCo environments with low-dimensional state inputs. Reported scores are averaged over 100 independent trials with 95% confidence intervals. Parametric policies use Multi-Layer Perceptrons; diffusion-policy results are reported in Section 8.

Across the tested environments, DARP yields substantial improvements over standard behavior cloning. In robotic manipulation, the gains range from 15-25 percentage points, while the locomotion benchmarks exhibit large score improvements (Table 1 and Table 2). Purely non-parametric methods such as R&P and LWR perform poorly on these tasks; MRIL almost always improves over vanilla BC, but the strongest results throughout this suite are obtained by the DARP architecture.

Given the practical instantiation described in Section 2.4, we next evaluate whether DARP continues to scale when the input representation becomes substantially higher dimensional, as in image-based settings.



Figure 3: The RoboCasa “DrawerClose” task, in which the robotic arm must close the open drawer.



Figure 4: The Robosuite “Threading” task, in which the robotic arm must insert the needle-like implement into the small hole.

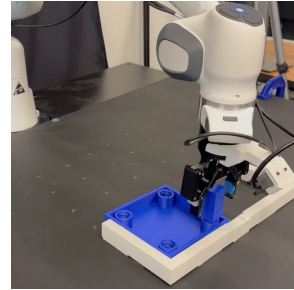


Figure 5: Real-world FurnitureBench square-table assembly task, in which the robot picks up a table leg and screws it into a hole at the tabletop corner.

3.3 CAN DARP HANDLE MORE COMPLEX STATE REPRESENTATION AND ACTION DISTRIBUTIONS? (Q2)

High-Dimensional Visual Input Representations. To assess the applicability of DARP beyond compact low-dimensional states, we evaluate it on simulated robotic-manipulation tasks using R3M image embeddings (Nair et al., 2022). This experiment tests whether the neighbor-based formulation remains effective when states are represented as high-dimensional feature vectors extracted

Method	Stack	Thrd.	Peg
BC	44	38	17
DARP	75	76	52

Table 3: Success rates (%) for vision-based Robosuite tasks.

Method	RoboCasa			Robosuite			Real
	Drawer	Door	Stove	Stack	Thrd.	Peg	Sq. Table
BC	54	29	28	47	37	46	44
DARP	85	45	43	72	63	62	92

Table 2: Comparison of DARP and BC across all three evaluation environments using low-dimensional state features. Scores denote success percentages (RoboCasa & Robosuite: 100 trials; Real: 50 trials). DARP consistently exceeds BC across tasks and environments, more than doubling BC’s real-world score. RoboCasa and Robosuite policies use Multi-Layer Perceptrons, whereas the real-world results use diffusion policy; see Section 8.

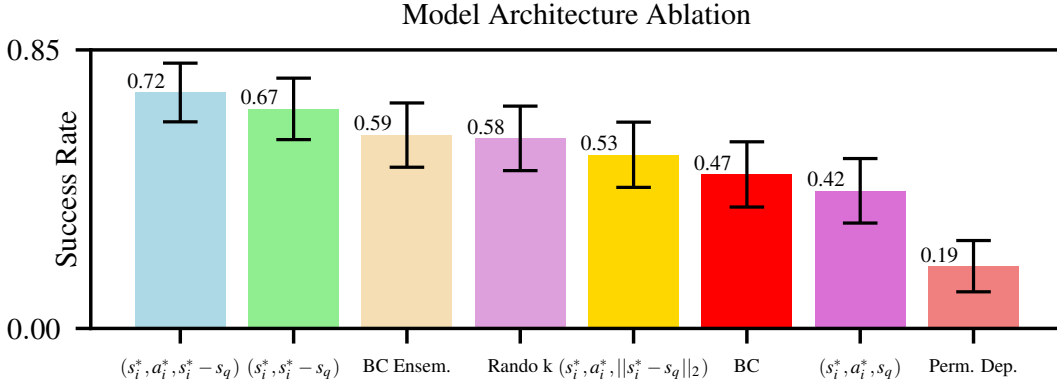


Figure 6: **Distance vectors and permutation invariance are major contributors to DARP’s success.** Ablating the core architecture shows that performance is most strongly associated with the difference-aware tuples $(s_i^*, a_i^*, s_i^* - s_q)$. Success rates are averaged over 100 trials on the Robosuite Stack environment with 95% confidence intervals.

from visual observations (Table 3). Notably, DARP not only outperforms standard BC, but its average improvement, $\sim 35\%$, exceeds the average improvement observed on low-dimensional Robosuite states ($\sim 22\%$). Empirically, this indicates that DARP adapts more effectively than standard BC to complex, high-dimensional state representations.

Multi-modal Action Distributions. We further show that DARP can solve complex multimodal imitation-learning tasks, including the Push-T environment, with more than 20% improvement over behavior cloning. Details are provided in Appendix A.2.2.

3.4 HOW DO DIFFERENT ARCHITECTURAL COMPONENTS CONTRIBUTE TO DARP’S PERFORMANCE GAINS? (Q3)

Ablation Study: To identify which architectural elements most strongly contribute to DARP’s gains, we conduct a systematic ablation over the principal design choices: (1) standard DARP; (2) DARP without neighbor actions; (3) an ensemble of 10 BC agents; (4) DARP with random rather than metric-based neighbors; (5) DARP using the L_2 norm of the distance vector; (6) the BC baseline, which receives only the query state s_q ; (7) DARP with the query state in place of the difference vector between query and neighbor states; and (8) DARP with a permutation-dependent, hence **not** permutation-invariant, aggregator for combining all a ’s. Figure 6 reports the results of this ablation.

The ablation study indicates that distance vectors and permutation invariance are essential to DARP’s performance, whereas neighbor actions contribute more modestly. Random neighbor selection performs poorly, confirming that meaningful distance metrics are crucial for constructing useful neighborhoods. The permutation-invariant aggregation function g is likewise critical, since permutation-dependent alternatives substantially degrade performance.

Divergence Analysis: To better characterize why DARP succeeds where standard BC fails, we analyze rollout pairs in which BC fails but DARP completes the task. We define the “step of diver-

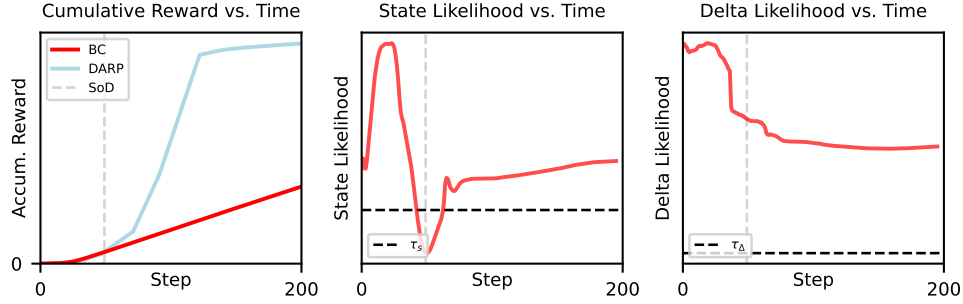


Figure 7: **Cumulative rewards for BC and DARP on the Robosuite Stack task show initially identical rollouts that diverge when BC fails and DARP succeeds.** The vertical dashed line marks the step of divergence, labeled “SoD”. At the SoD, the state likelihood is $< \tau_s$ (OOD), whereas the delta likelihood remains $> \tau_\Delta$ (in distribution).

gence” as the time step at which the reward trajectories of DARP and BC begin to differ substantially. We define τ_s and τ_Δ as the 1st percentile of likelihoods under the training set; equivalently, 1% of training-time deltas have likelihood below τ_Δ . Across all six rollouts spanning two tasks, Robosuite Stack and MuJoCo Hopper, the query state at the SoD has state likelihood $< \tau_s$ but delta likelihood $> \tau_\Delta$. This pattern supports the hypothesis that DARP’s gains arise partly from improved prediction on mildly out-of-distribution states, enabled by reparameterizing the problem through difference vectors to neighbors. Figure 7 shows reward drift, SoDs, and state and delta likelihoods for one task; Appendix A.2.4 provides additional robustness experiments.

4 RELATED WORK

Non-Parametric Imitation Learning Methods: Non-parametric IL algorithms can achieve unexpectedly strong performance by exploiting local structure in the demonstration distribution. VINN Pari et al. (2022) studies locally weighted regression for imitation learning and reports strong results in image-embedding spaces, while MiDiGaP von Hartz et al. (2025) uses mixtures of Gaussian processes to model multimodal trajectories and obtain rapid generalization. SEABO Lyu et al. (2024) uses retrieval to support offline RL by rewarding transitions that remain close to retrieved neighbors, thereby inducing a reward function. FlowRetrieval Lin et al. (2024), STRAP Memmel et al. (2025), and Behavior Retrieval Du et al. (2023) perform non-parametric retrieval and finetuning from large unlabeled datasets, enabling generalization through test-time training. DARP differs from these approaches through its distinctive parameterization of retrieved states as $(s_i^*, a_i^*, s_i^* - s_q)$ tuples and through its use of a learned semi-parametric policy rather than purely non-parametric aggregation or test-time adaptation. This design combines the variance reduction associated with local methods with the generalization capacity of parametric policies.

Smoothness-Constrained Policy Learning: A growing literature has examined explicit smoothness constraints as a mechanism for improving policy stability and robustness. L2C2 Kobayashi (2022) studies model-free RL under local Lipschitz-continuity constraints, achieving smoothness and noise robustness without sacrificing expressiveness, while Asadi et al. (2018) proposed a related methodology for model-based RL with Lipschitz-constrained models. CCIL Ke et al. (2024a) extends these ideas to imitation learning by using a Lipschitz-constrained dynamics model to generate synthetic corrective labels. Similar principles have also been scaled to humanoid controllers Chen et al. (2024) to reduce deployment-time shakiness. DARP differs from this line of work by inducing smoothness implicitly through an architectural reparameterization while retaining standard imitation-learning objectives.

In-Context Learning Methods: Recent work has also considered non-parametric retrieval through the lens of in-context imitation learning. REGENT Sridhar et al. (2025) investigates retrieval-augmented generalization by incorporating retrieved states, actions, and rewards into a causal transformer, whereas DPT Lee et al. (2023) uses supervised transformer pretraining to predict actions from query states and in-context datasets, effectively learning how to explore. Other in-context architectures include ICRT Fu et al. (2024), Instant Policy Vosylius & Johns (2025), and KAT Di Palo & Johns (2024). These methods primarily target rapid adaptation to new tasks and environments,

whereas DARP focuses on improving performance and stability within standard imitation-learning settings.

5 CONCLUSION

In this work, we introduced DARP, a semi-parametric retrieval-based imitation-learning method that reformulates behavior cloning around local difference structure rather than direct global state-to-action regression. We showed that the local-consistency principle underlying DARP corresponds to manifold regularization, or Laplacian smoothing, and that this structure can be incorporated implicitly through the policy architecture without modifying the standard imitation-learning objective. Empirically, DARP improves over standard BC across continuous-control, robotic-manipulation, visual-representation, multimodal-action, and real-world settings, with ablations indicating that difference vectors and permutation-invariant aggregation are central to the observed gains. These results suggest that reusing the demonstration corpus at inference time, when coupled with an appropriate difference-aware parameterization, provides a practical path toward more stable and robust imitation policies under distribution shift.

6 REPRODUCIBILITY STATEMENT

Upon acceptance, we will release supplementary source code, policy and environment configuration files, MuJoCo experiment data, scripts for generating expert demonstrations for Robosuite tasks through MimicGen, and modality-transformation code. Results are expected to match those reported in the paper on NVIDIA L40 and L40s GPUs, except for experiments requiring transformer-based components (REGENT and Set Transformer), which are non-deterministic and may therefore exhibit slight numerical variation relative to the reported values.

REFERENCES

- Kavosh Asadi, Dipendra Misra, and Michael Littman. Lipschitz continuity in model-based reinforcement learning. In *International Conference on Machine Learning*. PMLR, PMLR, 2018.
- Mikhail Belkin and Partha Niyogi. Towards a theoretical foundation for laplacian-based manifold methods. *Journal of Computer and System Sciences*, 74(8):1289–1308, 2008.
- Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Lucy Xiaoyang Shi, James Tanner, Quan Vuong, Anna Walling, Haohuan Wang, and Ury Zhilinsky. π_0 : A vision-language-action flow model for general robot control, 2024. URL <https://arxiv.org/abs/2410.24164>.
- Zixuan Chen, Xialin He, Yen-Jen Wang, Qiayuan Liao, Yanjie Ze, Zhongyu Li, S. Shankar Sastry, Jiajun Wu, Koushil Sreenath, Saurabh Gupta, and Xue Bin Peng. Learning smooth humanoid locomotion through lipschitz-constrained policies. *arxiv preprint arXiv:2410.11825*, 2024.
- Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 2024.
- Fan R. K. Chung. *Spectral Graph Theory*, volume 92 of *CBMS Regional Conference Series in Mathematics*. American Mathematical Society, 1997. doi: 10.1090/cbms/092.
- Michael Jae-Yoon Chung, Maxwell Forbes, Maya Cakmak, and Rajesh PN Rao. Accelerating imitation learning through crowdsourcing. In *2014 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4777–4784. IEEE, 2014.
- Norman Di Palo and Edward Johns. Keypoint action tokens enable in-context imitation learning in robotics. In *Proceedings of Robotics: Science and Systems (RSS)*, 2024.

-
- Maximilian Du, Suraj Nair, Dorsa Sadigh, and Chelsea Finn. Behavior retrieval: Few-shot imitation learning by querying unlabeled datasets. In *Proceedings of Robotics: Science and Systems (RSS)*, 2023.
- Justin Fu, Aviral Kumar, Ofir Nachum, George Tucker, and Sergey Levine. D4rl: Datasets for deep data-driven reinforcement learning, 2020.
- Letian Fu, Huang Huang, Gaurav Datta, Lawrence Yunliang Chen, William Chung-Ho Panitch, Fangchen Liu, Hui Li, and Ken Goldberg. In-context imitation learning via next-token prediction. *arXiv preprint arXiv:2408.15980*, 2024.
- Liyiming Ke, Yunchu Zhang, Abhay Deshpande, Siddhartha Srinivasa, and Abhishek Gupta. Ccil: Continuity-based data augmentation for corrective imitation learning. In *The Twelfth International Conference on Learning Representations*, 2024a. URL <https://openreview.net/forum?id=LQ6LQ8f4y8>.
- Liyiming Ke, Yunchu Zhang, Abhay Deshpande, Siddhartha S. Srinivasa, and Abhishek Gupta. CCIL: continuity-based data augmentation for corrective imitation learning. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024b. URL <https://openreview.net/forum?id=LQ6LQ8f4y8>.
- Taisuke Kobayashi. L2c2: Locally lipschitz continuous constraint towards stable and smooth reinforcement learning. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 4032–4039. IEEE, IEEE, October 2022.
- Jonathan Lee et al. Supervised pretraining can learn in-context reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, pp. 43057–43083, 2023.
- Juho Lee, Yoonho Lee, Jungtaek Kim, Adam Kosior, Seungjin Choi, and Yee Whye Teh. Set transformer: A framework for attention-based permutation-invariant neural networks. In *Proceedings of the 36th International Conference on Machine Learning*, pp. 3744–3753, 2019.
- Sergey Levine, Aviral Kumar, George Tucker, and Justin Fu. Offline reinforcement learning: Tutorial, review, and perspectives on open problems, 2020. URL <https://arxiv.org/abs/2005.01643>.
- Li-Heng Lin, Yuchen Cui, Amber Xie, Tianyu Hua, and Dorsa Sadigh. Flowretrieval: Flow-guided data retrieval for few-shot imitation learning. In Pulkrit Agrawal, Oliver Kroemer, and Wolfram Burgard (eds.), *Conference on Robot Learning, 6-9 November 2024, Munich, Germany*, volume 270 of *Proceedings of Machine Learning Research*, pp. 4084–4099. PMLR, 2024. URL <https://proceedings.mlr.press/v270/lin25a.html>.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR) 2023*, 2023. URL <https://arxiv.org/abs/2210.02747>. Originally appeared as arXiv preprint arXiv:2210.02747.
- Jiafei Lyu, Xiaoteng Ma, Le Wan, Runze Liu, Xiu Li, , and Zongqing Lu. Seabo: A simple search-based method for offline imitation learning. In *International Conference on Learning Representations*, 2024.
- Ajay Mandlekar, Soroush Nasiriany, Bowen Wen, Ireteyio Akinola, Yashraj Narang, Linxi Fan, Yuke Zhu, and Dieter Fox. Mimicgen: A data generation system for scalable robot learning using human demonstrations. In *7th Annual Conference on Robot Learning*, 2023.
- Elman Mansimov and Kyunghyun Cho. Simple nearest neighbor policy method for continuous control tasks. In *International Conference on Learning Representations (ICLR) 2018*, 2018. Under review as a conference paper at ICLR 2018.
- Marius Memmel, Jacob Berg, Bingqing Chen, Abhishek Gupta, and Jonathan Francis. Strap: Robot sub-trajectory retrieval for augmented policy learning. In *The Thirteenth International Conference on Learning Representations*, 2025.

- Suraj Nair, Aravind Rajeswaran, Vikash Kumar, Chelsea Finn, and Abhinav Gupta. R3M: A universal visual representation for robot manipulation. In Karen Liu, Dana Kulic, and Jeffrey Ichnowski (eds.), *Conference on Robot Learning, CoRL 2022, 14-18 December 2022, Auckland, New Zealand*, volume 205 of *Proceedings of Machine Learning Research*, pp. 892–909. PMLR, 2022. URL <https://proceedings.mlr.press/v205/nair23a.html>.
- Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. Robocasa: Large-scale simulation of everyday tasks for generalist robots. In *Robotics: Science and Systems (RSS)*, 2024.
- Jyothish Pari, Nur Muhammad Shafullah, Sridhar Pandian Arunachalam, and Lerrel Pinto. The surprising effectiveness of representation learning for visual imitation. In *Robotics: Science and Systems (RSS)*. Robotics: Science and Systems Foundation, June 2022. doi: 10.15607/RSS.2022.XVIII-052.
- Emmanuel Pignat and Sylvain Calinon. Bayesian gaussian mixture model for robotic policy imitation. *IEEE Robotics and Automation Letters*, 2019. URL <https://arxiv.org/pdf/1904.10716.pdf>. Preprint version.
- Dean Pomerleau. Efficient training of artificial neural networks for autonomous navigation. *Neural Comput.*, 3(1):88–97, 1991. doi: 10.1162/NECO.1991.3.1.88. URL <https://doi.org/10.1162/neco.1991.3.1.88>.
- Stéphane Ross and Drew Bagnell. Efficient reductions for imitation learning. In *Proceedings of the thirteenth international conference on artificial intelligence and statistics*, pp. 661–668. JMLR Workshop and Conference Proceedings, 2010.
- Stéphane Ross, Geoffrey J. Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In Geoffrey J. Gordon, David B. Dunson, and Miroslav Dudík (eds.), *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics, AISTATS 2011, Fort Lauderdale, USA, April 11-13, 2011*, volume 15 of *JMLR Proceedings*, pp. 627–635. JMLR.org, 2011. URL <http://proceedings.mlr.press/v15/ross11a/ross11a.pdf>.
- Steven L. Salzberg and David W. Aha. Learning to catch: Applying nearest neighbor algorithms to dynamic control tasks. In P. Cheeseman and R. W. Oldford (eds.), *Selecting Models from Data: Artificial Intelligence and Statistics IV*, volume 89 of *Lecture Notes in Statistics*, pp. 321–328. Springer, 1994. doi: 10.1007/978-1-4612-2660-4_33. URL https://link.springer.com/chapter/10.1007/978-1-4612-2660-4_33.
- Jonathan C. Spencer, Sanjiban Choudhury, Arun Venkatraman, Brian D. Ziebart, and J. Andrew Bagnell. Feedback in imitation learning: The three regimes of covariate shift. *CoRR*, abs/2102.02872, 2021. URL <https://arxiv.org/abs/2102.02872>.
- Kaustubh Sridhar, Souradeep Dutta, Dinesh Jayaraman, and Insup Lee. REGENT: A retrieval-augmented generalist agent that can act in-context in new environments. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025. Oral presentation.
- Emanuel Todorov, Tom Erez, and Yuval Tassa. Mujoco: A physics engine for model-based control. In *2012 IEEE/RSJ international conference on intelligent robots and systems*, pp. 5026–5033. IEEE, 2012.
- Arun Venkatraman, Martial Hebert, and J. Andrew Bagnell. Improving multi-step prediction of learned time series models. In Blai Bonet and Sven Koenig (eds.), *Proceedings of the Twenty-Ninth AAAI Conference on Artificial Intelligence, January 25-30, 2015, Austin, Texas, USA*, pp. 3024–3030. AAAI Press, 2015. doi: 10.1609/AAAI.V29I1.9590. URL <https://doi.org/10.1609/aaai.v29i1.9590>.
- Jan Ole von Hartz, Adrian Röfer, Joschka Boedecker, and Abhinav Valada. The unreasonable effectiveness of discrete-time gaussian process mixtures for robot policy learning. *arXiv preprint arXiv:2505.03296*, 2025.

756 Vitalis Vosylius and Edward Johns. Instant policy: In-context imitation learning via graph diffusion.
757 In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.
758

759 Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan R Salakhutdinov, and
760 Alexander J Smola. Deep sets. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus,
761 S. Vishwanathan, and R. Garnett (eds.), *Advances in Neural Information Processing Systems 30*,
762 pp. 3391–3401. Curran Associates, Inc., 2017. URL [http://papers.nips.cc/paper/](http://papers.nips.cc/paper/6931-deep-sets.pdf)
763 [6931-deep-sets.pdf](http://papers.nips.cc/paper/6931-deep-sets.pdf).

764 Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual
765 manipulation with low-cost hardware. In *Proceedings of Robotics: Science and Systems (RSS)*,
766 2023.

767 Dengyong Zhou, Olivier Bousquet, Thomas Navin Lal, Jason Weston, and Bernhard Schölkopf.
768 Learning with local and global consistency. In *Advances in Neural Information Processing Systems*
769 *(NeurIPS)*, volume 16, 2003.
770

771 Yuke Zhu, Josiah Wong, Ajay Mandlekar, Roberto Martín-Martín, Abhishek Joshi, Soroush Nasiriany,
772 Yifeng Zhu, and Kevin Lin. robosuite: A modular simulation framework and benchmark for robot
773 learning. In *arXiv preprint arXiv:2009.12293*, 2020.
774
775
776
777
778
779
780
781
782
783
784
785
786
787
788
789
790
791
792
793
794
795
796
797
798
799
800
801
802
803
804
805
806
807
808
809

A APPENDIX

A.1 LEMMAS AND PROOFS

A.1.1 PROOF OF THEOREM 1

To prove Theorem 1, we begin from a standard result in graph-based semi-supervised learning and manifold regularization (Chung, 1997; Zhou et al., 2003; Belkin & Niyogi, 2008).

Lemma 1 (Smoothness regularizer as k -NN graph Laplacian penalty). *Let $\{s_1^*, \dots, s_n^*\}$ be the expert states with corresponding predicted actions $f(s_i^*) \in \mathbb{R}^{d_a}$. For each i , let $\mathcal{N}_k(s_i^*)$ denote the indices of the k -nearest neighbors of s_i^* (excluding i). Define asymmetric weights*

$$\tilde{W}_{ij} = \begin{cases} w_j(s_i^*), & \text{if } j \in \mathcal{N}_k(s_i^*), \\ 0, & \text{otherwise,} \end{cases}$$

and construct a symmetric affinity matrix

$$W_{ij} = \frac{1}{2}(\tilde{W}_{ij} + \tilde{W}_{ji}).$$

Let D be the degree matrix with $D_{ii} = \sum_j W_{ij}$, and define the k -NN graph Laplacian $L = D - W$.

Then the smoothness regularizer can be written as the quadratic form

$$\mathcal{L}_S(f) = \frac{1}{n} \sum_{i=1}^n \sum_{j \in \mathcal{N}_k(s_i^*)} w_j(s_i^*) \|f(s_i^*) - f(s_j^*)\|^2 \propto \text{Tr}(F^\top L F),$$

where $F = [f(s_1^*), f(s_2^*), \dots, f(s_n^*)]^\top \in \mathbb{R}^{n \times d_a}$. Equivalently, in the scalar case,

$$\mathcal{L}_S(f) \propto f^\top L f.$$

Corollary 1 (Continuum limit of smoothness regularizer). *Assume states $\{s_i^*\}_{i=1}^n$ are sampled i.i.d. from a smooth density $p(s^*)$ supported on an m -dimensional C^2 manifold $\mathcal{M} \subset \mathbb{R}^d$. Let W be the symmetrized k -NN affinity matrix constructed from a kernel K_Δ with bandwidth h , and let $L = D - W$ be the graph Laplacian.*

If $n \rightarrow \infty$, $h \rightarrow 0$, and $nh^{m+2} \rightarrow \infty$, then the normalized quadratic form converges to the weighted Dirichlet energy:

$$\frac{1}{n^2 h^{m+2}} \text{Tr}(F^\top L F) \rightarrow C_K \int_{\mathcal{M}} \|\nabla_{\mathcal{M}} f(s)\|_2^2 p(s)^2 d\text{vol}(s),$$

where $C_K > 0$ is a constant depending only on the kernel K_Δ .

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ be the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$. Let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider two estimators trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Then, under the smoothness assumption on f , the following hold:

- (i) Variance reduction: The Laplacian penalty in MRIL acts as a data-dependent Tikhonov regularizer, yielding smaller estimator variance than vanilla BC.

(ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants with arbitrarily large Lipschitz constants between training states.

(iii) Policy stability: In a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^4$$

accumulates error linearly for vanilla BC, but sublinearly for MRIL, since the smoothness regularizer enforces $\|f(s^*) - f(s'^*)\| = O(\|s^* - s'^*\|)$ for neighbors s^*, s'^* .

This suggests that MRIL enjoys strictly better generalization and stability guarantees than BC.

Proof. We establish the three claims sequentially.

(i) Variance reduction. The Laplacian penalty in $\mathcal{L}_{\text{MRIL}}$ is

$$\sum_{i,j} W_{ij} \|f(s_i^*) - f(s_j^*)\|^2 = 2f^\top Lf,$$

where $L = D - W$ is the graph Laplacian and W is the k -NN affinity matrix. By Lemma 1, this expression is precisely the empirical Dirichlet energy of f on the k -NN graph. It is well known (Zhou et al., 2003; Belkin & Niyogi, 2008) that such a quadratic penalty is equivalent to Tikhonov regularization with respect to the graph-Laplacian norm $\|f\|_L^2 = f^\top Lf$. Within statistical learning theory, adding a Tikhonov penalty reduces estimator variance relative to the unregularized solution while keeping the bias term controlled. Consequently, MRIL has smaller estimator variance than vanilla BC, which contains no analogous penalty.

(ii) Smoothness guarantee. Consider the continuum limit (Corollary 1): for i.i.d. samples $\{s_i^*\}$ from density p on a smooth manifold $\mathcal{M}$, the normalized penalty converges to

$$\int_{\mathcal{M}} \|\nabla f(s)\|^2 p(s)^2 d\text{vol}(s).$$

This quantity is the weighted Dirichlet energy of f on $\mathcal{M}$. When the integral is finite, f belongs to the Sobolev space $H^1(\mathcal{M}, p^2)$, and in particular f is locally Lipschitz almost everywhere with

$$\|f(s^*) - f(s'^*)\| \leq C\|s^* - s'^*\| \quad \text{for } p\text{-a.e. neighbor pairs } s^*, s'^*.$$

Thus minimizers of $\mathcal{L}_{\text{MRIL}}$ possess uniformly bounded local Lipschitz constants along high-density regions of the state space. By contrast, vanilla BC imposes no comparable constraint: any oscillatory interpolant that matches the training data exactly obtains the same supervised risk, so arbitrarily large Lipschitz constants remain admissible.

(iii) Policy stability. Let $\Delta_t = \|s_t - s_t^*\|$ denote the deviation at time t . Under Lipschitz transition dynamics T ,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|.$$

Decompose the action error:

$$\|f(s_t) - f^*(s_t^*)\| \leq \|f(s_t) - f^*(s_t)\| + \|f^*(s_t) - f^*(s_t^*)\|.$$

For vanilla BC, the first term $\|f(s_t) - f^*(s_t)\|$ is minimized only on the empirical distribution $P_{\mathcal{S}}$; away from that distribution, it may be $O(1)$ regardless of Δ_t . The second term satisfies $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Hence the recursion can take the form

$$\Delta_{t+1} \leq L_s \Delta_t + L_a (O(1) + O(\Delta_t)),$$

which accumulates linearly in t .

For MRIL, the Laplacian penalty enforces

$$\|f(s^*) - f(s'^*)\| \leq C\|s^* - s'^*\| \quad \text{for neighbor pairs } (s^*, s'^*),$$

⁴ L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

as shown in part (ii). Thus $\|f(s_t) - f^*(s_t)\| = O(h^2)$ (where h is the radius of the kernel bandwidth around the point s_t) by local-linear regression error bounds, and $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Combining these,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a (O(h^2) + O(\Delta_t)).$$

Because the constant multiplying Δ_t is strictly smaller under the smoothness constraint, cumulative error grows more slowly than under vanilla BC. In particular, for sufficiently small h , error growth is sublinear in the rollout horizon, whereas vanilla BC exhibits linear growth.

Conclusion. Claims (i)–(iii) together establish that MRIL provides lower variance, uniform smoothness control, and sublinear rollout-error accumulation relative to vanilla BC, completing the proof. $\square$

Kernel choice. For the IC smoothness regularizer, we use a Gaussian kernel

$$w_i(s^*) \propto \exp\left(-\frac{\|s^* - s_i\|^2}{2h^2}\right), \quad \sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) = 1,$$

with bandwidth h set to the median distance to the k -th nearest neighbor across the dataset. This convention is standard in manifold regularization (Belkin & Niyogi, 2008; Zhou et al., 2003) and ensures that the graph Laplacian penalty converges to the Dirichlet energy in the continuum limit. Empirically, this default was stable across tasks, although other kernels, such as uniform k -NN weighting or exponential decay, produced qualitatively similar results.

A.1.2 PROOF OF THEOREM 2

We begin with the lemmas needed to characterize the spectral forms of explicit Laplacian regularization and neighbor aggregation.

Lemma 2 (Spectral form of explicit Laplacian regularization). *Let L be the symmetric normalized graph Laplacian with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$. The minimizer of the penalized objective*

$$\mathcal{L}_\lambda(f) = \|f - a^*\|^2 + \lambda f^\top L f$$

has the closed-form expansion

$$f_\lambda = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j.$$

Thus λ directly determines the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$ applied to each Laplacian mode.

Proof. Diagonalize $L = U \Lambda U^\top$ with $U = [u_1, \dots, u_n]$ orthogonal and $\Lambda = \text{diag}(\mu_1, \dots, \mu_n)$. Write $f = U c$, $a^* = U b$ in this basis. The objective becomes

$$\|U c - U b\|^2 + \lambda c^\top \Lambda c = \sum_{j=1}^n (c_j - b_j)^2 + \lambda \mu_j c_j^2.$$

Minimizing each term yields $c_j = \frac{1}{1 + \lambda \mu_j} b_j$. Transforming back to the original basis yields the stated expansion. $\square$

Lemma 3 (Spectral form of neighbor aggregation). *Let $S = D^{-1}A$ be the random-walk matrix of the k -NN graph, with adjacency A and degree D . For any prediction vector f , the neighbor-averaged prediction is $\hat{f} = S f$. In the Laplacian eigenbasis, this corresponds to the spectral filter*

$$\hat{f} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j,$$

i.e. $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

Proof. By definition, $L = I - D^{-1/2} A D^{-1/2}$ and $S = D^{-1} A = I - L_{\text{rw}}$ where $L_{\text{rw}} = D^{-1} L$ is the random-walk Laplacian. Since L_{rw} and L share the same spectrum up to similarity transform, the eigenbasis $\{u_j\}$ diagonalizes S . Thus for each mode u_j , $S u_j = (1 - \mu_j) u_j$, yielding the claimed spectral filter. $\square$

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.

The minimizers of the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) have the following closed form expansions

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

iMRIL's neighbor-aggregation step applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , preserving low-frequency modes while attenuating high-frequency modes. The correspondence between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} therefore shows that iMRIL implements a built-in form of Laplacian smoothing (MRIL) with effective $\lambda \approx 1$ in normalized units. Unlike explicit regularization, this implicit filter introduces no additional hyperparameter to tune.

Proof. From Lemma 3, the neighbor aggregation operator $S = D^{-1}A$ acts on Laplacian eigenmodes u_j as

$$Su_j = (1 - \mu_j)u_j,$$

where μ_j are the normalized Laplacian eigenvalues. Thus in the graph Fourier basis, neighbor aggregation corresponds to multiplying each mode by the fixed spectral filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

On the other hand, Lemma 2 shows that explicit Laplacian regularization with parameter λ yields the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$. Both filters downweight high-frequency modes ($\mu \gg 0$) while preserving low-frequency modes ($\mu \approx 0$). The key difference is that $\phi_\lambda(\mu)$ requires tuning λ , whereas $\phi_{\text{DARP}}(\mu)$ is parameter-free.

To see the equivalence, note that for small μ ,

$$\phi_{\text{DARP}}(\mu) = 1 - \mu \approx (1 + \mu)^{-1} = \phi_{\lambda=1}(\mu) \quad \text{up to } O(\mu^2) \text{ terms.}$$

Thus DARP can be interpreted as performing Laplacian smoothing with an effective regularization weight of order $\lambda \approx 1$ in normalized units. Moreover, for large μ , $\phi_{\text{DARP}}(\mu)$ damps high-frequency modes even more strongly by driving them toward zero, providing a sharper low-pass effect than explicit regularization.

Therefore, DARP's aggregation step is mathematically equivalent to implicit Laplacian regularization with the fixed spectral filter ϕ_{DARP} , thereby eliminating the need to tune λ explicitly. $\square$

Thus, DARP may be interpreted as a form of *locally adaptive implicit regularization*: instead of introducing an explicit global coefficient λ , its neighbor-aggregation step enforces smoothness

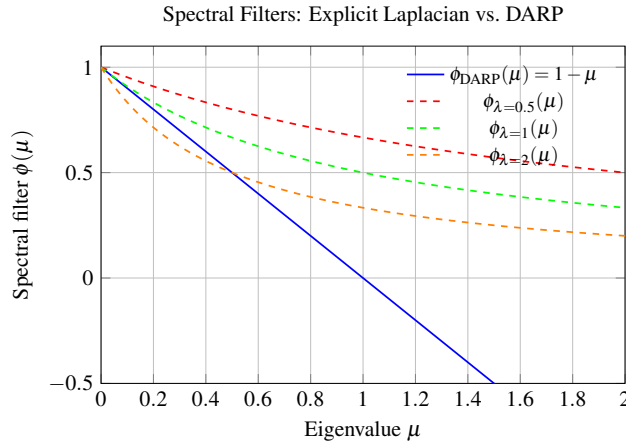


Figure 8: DARP achieves sharper low-pass filtering

automatically through the retrieval graph. The effective regularization strength varies with local degree and neighborhood geometry, thereby adapting to the density of the expert demonstrations. In spectral terms, this behavior corresponds to the fixed filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$, which attenuates high-frequency modes more aggressively than any fixed explicit λ . Figure 8 visualizes this comparison and shows that DARP obtains sharper low-pass filtering without hyperparameter tuning.

A.2 ADDITIONAL EXPERIMENTAL DETAILS

A.2.1 RETRIEVAL

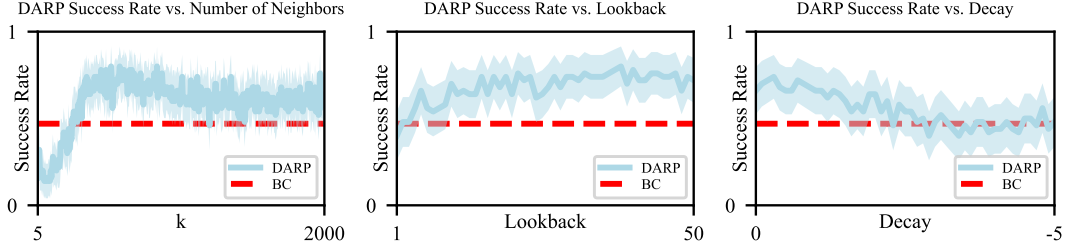


Figure 9: Retrieval-hyperparameter sensitivity for DARP: (left) performance is poor with very few neighbors, reaches a global optimum around 500 retrieved neighbors, and then plateaus slightly above BC as k approaches the dataset size; (center) performance generally improves modestly as additional history is incorporated and falls below BC only when little or no history is used; (right) performance depends on the weighting assigned to older observations during retrieval. When the decay is too large, DARP behaves similarly to a model with little or no lookback and performs worse than BC. Success rates are measured over 50 trials on the Robosuite Stack environment, with 95% confidence intervals.

The choice of distance function $d(s_q, s_i^*)$ used to select the k neighbors is consequential. Although simple Euclidean distance between states can be effective, our experiments use a modestly modified procedure that exploits the sequential structure of trajectories by incorporating history into the distance calculation. Consider a query trajectory $S_q = (\dots, s_{q,-1}, s_{q,0})$, where $s_{q,0}$ is the current query state s_q . To compute $d(s_q, s_i^*)$ for a dataset state s_i^* , we first identify the expert trajectory containing that state, denoted S_j^* , and the corresponding time index, denoted t , so that s_i^* can be written as $s_{j,t}^*$. For a lookback parameter ℓ specifying the number of past states to include, we define:

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\|$$

This expression accumulates Euclidean distances between the current and previous $\ell - 1$ states of the query and source trajectories, assuming all indices are valid. In practice, however, more recent states should usually exert greater influence over neighbor selection. Accordingly, for an exponential decay rate $\gamma \geq 0$, we use

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\| \cdot e^{-\gamma n}$$

Figure 9 reports how task success varies as these retrieval parameters are swept.

A.2.2 CAN DARP HANDLE TASKS REQUIRING THE REPRESENTATION OF MULTI-MODAL ACTION DISTRIBUTIONS?

We evaluate DARP’s capacity to model complex action distributions on the Push-T task from (Chi et al., 2024), which requires representing multimodal action structure (Figure 10). In this experiment, DARP uses a Set Transformer head to predict the parameters of a Gaussian Mixture Model. With this GMM head, DARP handles multimodal distributions effectively and improves over BC by 22% on Push-T (Q2), as shown in Table 4. This result demonstrates that DARP can be adapted to settings requiring multimodal action-distribution modeling.

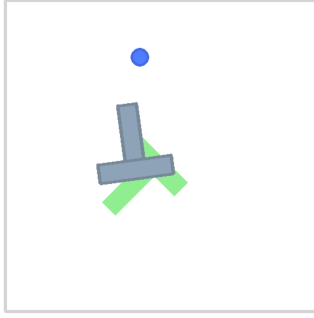


Figure 10: **Push-T Environment.** The agent controls the blue circle in order to push the T-shaped block.

Method	Score
BC	48 ± 8
DARP	70 ± 8

Table 4: **Push-T Results.** Averaged over 100 trials, DARP achieves higher scores than BC.

A.2.3 CAN DARP HANDLE DISCONTINUOUS ENVIRONMENTS WHERE NEARBY STATES MAY REQUIRE OPPOSING ACTIONS?

A central concern for neighbor-based approaches is their behavior in environments with sharp discontinuities, where states that are close under Euclidean distance may nevertheless require substantially different actions. To probe this failure mode, we construct a stress test using a modified version of D4RL’s Umaze environment, visualized in Figure 11.

Even in this deliberately challenging discontinuous environment, DARP achieves a 57% success rate compared with BC’s 25% (Q3; Table 5). This outcome suggests that distance vectors and permutation-invariant aggregation help the model distinguish appropriate from inappropriate neighbors, even when spatial proximity does not imply action similarity.

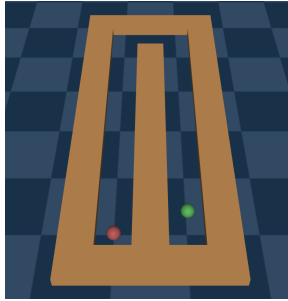


Figure 11: **Long maze environment.** The task requires moving a force-actuated ball from the green start location to the red destination.

Method	Succ. (%)
BC	25
DARP	57

Table 5: **Long maze results.** Averaged over 100 trials, DARP substantially outperforms BC.

A.2.4 CAN DARP RECOVER FROM BC ERROR?

To examine DARP’s robustness to accumulated error, we roll out a BC agent in an environment where it is known to fail and, every k steps, create a forked copy of the environment from which a DARP agent is rolled out. The results in Fig. 12 show that, even as BC approaches failure and drifts away from the support of expert demonstrations, DARP can recover and achieve high scores. This supports the conclusion that DARP is more robust to error accumulation and can operate effectively in the mildly out-of-distribution states visited by a failing BC policy.

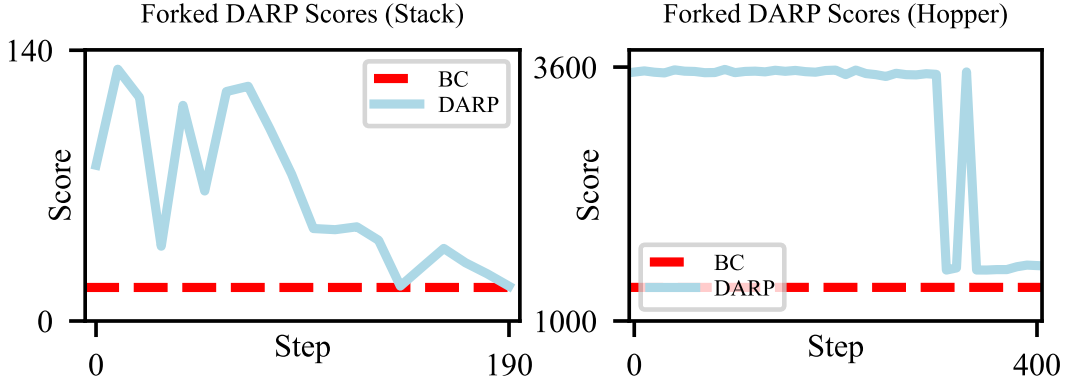


Figure 12: Across two tasks, Robosuite Stack and MuJoCo Hopper, we roll out a BC agent and fork the environment every k steps (here, $k = 10$). Even as BC approaches the terminal portion of a failing rollout, DARP can still recover to high scores, becoming unable to do so only roughly halfway through the Stack rollout and about 80% through the Hopper rollout.

Environment	Method	Improvement over BC (%)
Hopper	CCIL	32.6%
	DARP	53.2%
Walker	CCIL	84.1%
	DARP	84.1%
Ant	CCIL	12.6%
	DARP	84.5%
HalfCheetah	CCIL	5.4%
	DARP	418.7%

Table 6: **Relative improvement over BC compared with CCIL.** Comparison between DARP and CCIL on standard MuJoCo benchmarks. DARP achieves equal or higher relative gains across all tasks.

A.2.5 COMPARISON WITH CCIL

Table 6 compares our method with CCIL, a baseline that explicitly induces smoothness. Using the scores reported in the CCIL paper, we compare each method’s percentage improvement over BC. DARP substantially outperforms CCIL on three of four environments, with an especially large margin on HalfCheetah (418.7% versus 5.4% improvement).

A.2.6 DISTANCE METRIC SENSITIVITY

Method	Success Rate (%)
BC	47
DARP (Cosine Similarity)	70
DARP (Euclidean Distance)	75

Table 7: **Distance Metric Sensitivity.** Success-rate comparison using R3M features. DARP with Euclidean distance and cosine similarity perform similarly, exceeding BC by 28% and 23%, respectively. Results are reported on the Robosuite Stack task.

Although the main experiments use Euclidean distance to select nearby neighbors, alternative metrics such as cosine similarity are natural candidates, particularly in high-dimensional embedding spaces such as R3M. As shown in Table 7, DARP performs similarly with cosine similarity, only 5% below

the Euclidean-distance variant. This suggests that DARP is relatively robust to the choice of retrieval metric.

A.2.7 DARP IN COMBINATION WITH DIFFUSION POLICY

Method	Stack	Threading	Peg Insertion
BC (MLP)	34	12	28
DARP (MLP)	62	38	40
Diffusion	52	44	54
DARP w/ Diffusion	82	72	66

Table 8: **DARP in combination with diffusion policy.** DARP yields substantial improvements when applied to both standard MLP policies and diffusion policies. Results are success rates over 50 trials on Robosuite Stack, Threading, and Peg Insertion, using state-based inputs. Because these experiments use many fewer demonstrations than Table 2, the baseline BC and DARP values do not match those reported there.

Although the main reported experiments use an MLP backbone, diffusion policy (Chi et al., 2024) is a state-of-the-art model class for imitation learning, especially in manipulation settings. Table 8 shows that DARP is compatible with diffusion and can be combined with it to obtain an even stronger model. With a diffusion backbone, the DARP architecture improves over DARP with an MLP backbone by 20%, 34%, and 26% success-rate points on the three tasks, respectively, and exceeds standard BC by 48%, 60%, and 38% overall.

Method	Training (s/epoch)	Inference (s/step)
Diffusion	5.610	0.15700
DARP (MLP)	0.559	0.00437

Table 9: **Computational efficiency of DARP with an MLP backbone and diffusion policy.** Runtime-cost comparison between DARP (MLP) and diffusion policy. Diffusion policy is substantially more expensive, running $\approx 10\times$ slower during training and $\approx 36\times$ slower during inference.

We also find empirically that DARP with an MLP backbone is much faster than standard diffusion, especially at inference time, as shown in Table 9.

A.2.8 DARP TRAINED ON HUMAN DEMONSTRATIONS

Method	Success Rate (%)
BC	45
DARP	60

Table 10: **Results with human demonstrations.** Success-rate comparison when training on human data rather than data collected by RL policies. Results are reported on the Robosuite Stack task.

The expert demonstrations used for MuJoCo and Robosuite training are collected by an optimal reinforcement-learning policy. It is therefore important to verify that DARP retains its advantage over BC when the expert demonstrations are collected from humans. On human demonstrations for the Robosuite Stack task, DARP outperforms standard BC by 15%, as reported in Table 10.

A.2.9 CHOICE OF RETRIEVAL HYPERPARAMETERS

Environment	Method	Score (Mean)
Hopper	BC	507.49
	DARP	805.69
Stack	BC	0.12
	DARP	0.31

Table 11: **Performance comparison when validation loss selects training epochs and retrieval hyperparameters.** Mean scores on Hopper and Stack environments show that DARP maintains a performance advantage over BC.

Figure 9 shows that some retrieval-parameter choices, such as using very few neighbors, can cause DARP to underperform standard BC. However, selecting retrieval hyperparameters that minimize validation loss is an effective strategy for identifying performant settings, as shown in Table 11.

A.3 PSEUDOCODE

Algorithm 1 provides pseudocode for the DARP procedure.

Algorithm 1 Difference-Aware Retrieval Policies

```

1: Input: Expert demonstrations  $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$ , number of neighbors  $k$ 
2: Initialize:  $f$  parameters  $\theta$ 
3: if  $g$  is parametric then
4:   Initialize:  $g$  parameters  $\psi$ 
5: end if
6: // Training Loop
7: while not converged do
8:   Sample batch of query data  $(s_q^*, a_q^*) \sim \mathcal{D}^*$ 
9:   for each query pair  $(s_q^*, a_q^*)$  in batch do
10:    // Find  $k$ -Nearest Neighbors from the entire dataset  $\mathcal{D}^*$ 
11:     $\mathcal{N}_k(s_q^*) \leftarrow \text{argmin-}k; d(s_q^*, s_j^*)$ 
12:    // Compute Neighbor-based Predictions
13:    for each neighbor index  $i \in \mathcal{N}_k(s_q^*)$  do
14:       $a'_i \leftarrow f_\theta(s_i^*, a_i^*, s_i^* - s_q^*)$ 
15:    end for
16:    // Aggregate Predictions
17:    if  $g$  is parametric then
18:       $\hat{a}_q \leftarrow g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
19:    else
20:       $\hat{a}_q \leftarrow g(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
21:    end if
22:  end for
23:  // Update Parameters based on the batch loss
24:   $\mathcal{L} \leftarrow \sum_{(s_q^*, a_q^*) \in \text{batch}} \|\hat{a}_q - a_q^*\|^2$ 
25:  // Gradient descent step
26:   $\theta \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}$ 
27:  if  $g$  is parametric then
28:     $\psi \leftarrow \psi - \alpha \nabla_\psi \mathcal{L}$ 
29:  end if
30: end while
31: Output: Trained parameters  $\theta$  and, if applicable,  $\psi$ 

```

A.4 RUNTIME ANALYSIS

Environment	k	Train (s/epoch)	Test (s/step)
Hopper (1,000 datapoints)	BC	0.102	0.00127
	100	0.126	0.00413
	200	0.128	0.00418
	300	0.130	0.00420
	400	0.131	0.00421
	500	0.130	0.00419
Stack (4,200 datapoints)	BC	0.431	0.00139
	100	0.556	0.00437
	250	0.539	0.00442
	500	0.559	0.00437
	750	0.576	0.00433
	1000	0.591	0.00478
	1500	0.682	0.00452
	2000	0.758	0.00451

Table 12: **Runtime comparison across environments.** Training and testing speeds, measured in seconds, for Behavior Cloning (BC) and multiple values of k on Hopper and Stack datasets.

Table 12 analyzes the computational overhead of DARP during both training and inference. The results indicate that computational cost scales sub-linearly as k increases and that inference remains tractable for real-time robotic control. For example, on the Robosuite Stack task with $k = 500$, inference requires approximately 0.00437 seconds per step on our hardware, corresponding to a control frequency above 230 Hz.